# Double Pendulum Report: Abstract

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In this report, we study different numerical integration techniques namely Euler's method, Runge-Kutta second order, and Runge-Kutta fourth order. We then apply Runge-Kutta fourth order to study the double pendulum system, which exhibits chaotic behavior. We specifically explore the period T it takes the double pendulum to flip as a function of the initial conditions $\theta^1(t=0)$ - the angle between the first section and the vertical - and $\theta^2(t=0)$ - the angle between the second section and the vertical. A double pendulum is considered to have "flipped" if the bottom portion does a full rotation in either direction - i.e. $\theta^2(t) > 2\pi$ or $\theta^2(t) < -2\pi$. We perform 360,000 different simulations using a $(\theta^1_i(t=0), \theta^2_j(t=0))$ $i, j = 1, 2, \ldots 600$, where $\theta^1_i(t=0) = -\pi + 2\pi * (i-1)/599$, $\theta^2_j(t=0) = -\pi + 2\pi * (j-1)/599$. We generally observe that the simulations that started at greater angles to the vertical flipped, and those that started closer to the vertical did not.